

A Fixed-Rank Calkin Obstruction to Zero-Noise Quotient Contraction

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Abstract

RH-269 asks for a contractive power of a limiting orthogonal quotient. We show that this cannot occur at the deterministic zero-noise endpoint under any fixed-rank removal in the natural stationary L^2 geometry. The deterministic Koopman limit is an isometry on an infinite-dimensional Perron/parity complement, and finite-rank compression leaves its Calkin class unchanged. After Hardy scaling every quotient power has norm at least $r_H^{-m} > 1$. Rank-growing selectors and growing block depths are not excluded.

1 Calkin theorem

Let U_0 be the deterministic Koopman isometry on the invariant complement of the constant and component-sign functions. Let P be any finite-rank orthogonal projection and $Q = I - P$.

Theorem 1.1 (fixed-rank quotient no-go). *For every $m \geq 1$,*

$$\|(QU_0Q)^m\| \geq 1, \quad \|\{r_H^{-1}QU_0Q\}^m\| \geq r_H^{-m} > 1.$$

Thus no fixed-rank limiting quotient in this geometry has a contractive power after Hardy scaling.

Proof. In the Calkin algebra finite-rank operators vanish, so $[QU_0Q] = [U_0]$. Since $U_0^*U_0 = I$, the class $[U_0]$ is an isometry and $\|[U_0]^m\| = 1$. The quotient norm is bounded above by the operator norm, which gives the first inequality. Scalar Hardy scaling gives the second. $\square$

2 Rank firewall

A common isolating contour in the RH-269 setting gives norm-continuous Riesz projections and hence a locally constant finite rank. The RH-222 frozen cloud ranks, including Perron and parity, run from 6 to 37. Therefore a single fixed-rank contour cannot encode that rank-growing atlas. This is a logical rank mismatch, not a continuum theorem about the stored roots.

Remark 2.1. The theorem does not exclude $P = P_\sigma$ with rank tending to infinity, $m = m(\sigma) \rightarrow \infty$, or an anisotropic space whose zero-noise limit is not the stationary Koopman isometry.

3 Boundary

Together with RH-276, the result closes the raw fixed-rank zero-noise $\mathcal{S}_2$ /contraction branch. It does not close the rank-growing trace route, Gate A, or any later Gate.